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2027 CYCLE

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CUET UG

Economics

Production and Costs

Official Syllabus · Course I: Introductory Microeconomics · Class 12 · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Economics syllabus 2026, the current 2027 baseline, Course I)	Skills the questions test
The production function	Production function: short run and long run production function; shapes of TP, MP and AP curves	Telling short run from long run by factor fixity, computing TP, MP and AP, and reading the three phases.
Costs	Concepts of costs; short-run cost curves (fixed and variable costs; total, average and marginal costs)	Splitting fixed from variable cost, building a full cost table, and reading the U-shapes with the MC crossings.
Revenue	Concepts of revenue — total, average and marginal revenue — and their relationship	Linking TR, AR and MR for a price-taking firm and recognising the horizontal price line.
Producer's equilibrium	Producer's equilibrium with the help of MC and MR	Applying MC = MR with the rising-MC condition and ruling out false equilibria.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Economics paper follows NCERT Class 12 Economics, and this unit feeds the next two — supply behaviour (firm under perfect competition) and price determination (market equilibrium) both run on the cost and revenue machinery built here.

The Production Function: Two Time Horizons

A bakery cannot conjure a second oven overnight, but it CAN call two more workers tonight. That everyday gap — some things change slowly, others instantly — is exactly what economists mean by the short run and the long run. The **production function** is the rulebook: it tells you the MAXIMUM output each combination of inputs can produce with today's technology.

Feature	Short run	Long run
Which factors can change?	At least ONE factor stays fixed — land, factory, machines	ALL factors are variable — even the factory size
How output changes	Add or remove workers on the SAME plant	Change both workers AND the plant
Named law	Law of variable proportions	Returns to scale
What it is NOT	Not a fixed number of days or months	Not "many years" — it depends on the industry

- **The long run is not a calendar.** A tea stall's long run might be a week; an airline's might be a decade. The split is about **which factors are variable**, never about dates.
- **The three product measures** (all about the variable factor, labour):
  - **TP (Total Product)** — total output from all workers together.
  - **MP (Marginal Product)** — the EXTRA output when ONE more worker joins: $\text{MP} = \frac{\Delta\ \text{TP}}{\Delta L}$
  - **AP (Average Product)** — output per worker: $\text{AP} = \text{TP}\ /L$
- **Tiny example:** 3 workers make 36 cups of tea, a 4th worker lifts TP to 48. The 4th worker's MP is 12 cups; AP is now 48/4 = 12 cups.

THE THREE PRODUCT FORMULAS

$\text{MP} = \frac{\Delta\ \text{TP}}{\Delta L}$ $\text{AP} = \frac{\text{TP}}{L}$ TU of production: $\text{TP} = \Sigma\ \text{MP}$

Letters: TP = total product · MP = marginal product of the next worker · AP = average product per worker · L = number of workers.

Sum rule: TP is the RUNNING TOTAL of MPs — the MP of the 4th worker is TP(4) − TP(3).

SEVEN WORKERS ON ONE PLOT — THE FULL TP/MP/AP TABLE

Fixed land, workers added one by one:

Workers (L)	TP (kg)	MP (kg)	AP (kg)	What is happening
1	8	8	8	First worker does everything
2	20	12	10	Specialisation kicks in — MP rises
3	36	16	12	MP peaks — best team size for speed

Workers (L)	TP (kg)	MP (kg)	AP (kg)	What is happening
4	48	12	12	MP = AP exactly: AP at its MAXIMUM
5	56	8	11.2	MP falls below AP — AP starts sliding
6	60	4	10	Still adds output, but barely
7	58	−2	8.3	Overcrowded — TP actually FALLS

- **Check the sum rule:** TP at 4 workers = 8 + 12 + 16 + 12 = 48 ✓
- **Check the crossing:** MP equals AP (12 = 12) exactly at AP's maximum — 4 workers. Before it, MP pulls AP up; after it, MP drags AP down.

The Law of Variable Proportions

Keep land fixed and add workers: output first surges, then crawls, then can actually shrink. This pattern — the **law of variable proportions** — is the short run's story, and it splits neatly into three phases you must be able to mark on a diagram.

- **Phase I — increasing returns:** TP rises at an INCREASING rate. MP rises (more workers = better teamwork and specialisation), and AP keeps rising because MP sits above it.
- **Phase II — diminishing returns:** TP rises at a DIMINISHING rate. MP falls but stays positive; AP now falls too (MP is below it). **Every rational producer operates here** — Phase I wastes fixed capacity, Phase III destroys output.
- **Phase III — negative returns:** TP FALLS. MP goes negative: the 7th worker on one tiny plot knocks over more than he produces. Too many hands, too little land.
- **Why diminishing returns happen:** the FIXED factor gets crowded. Each worker now shares the same land, same oven, same tools — extra workers contribute less and less. Factor proportions go unbalanced.

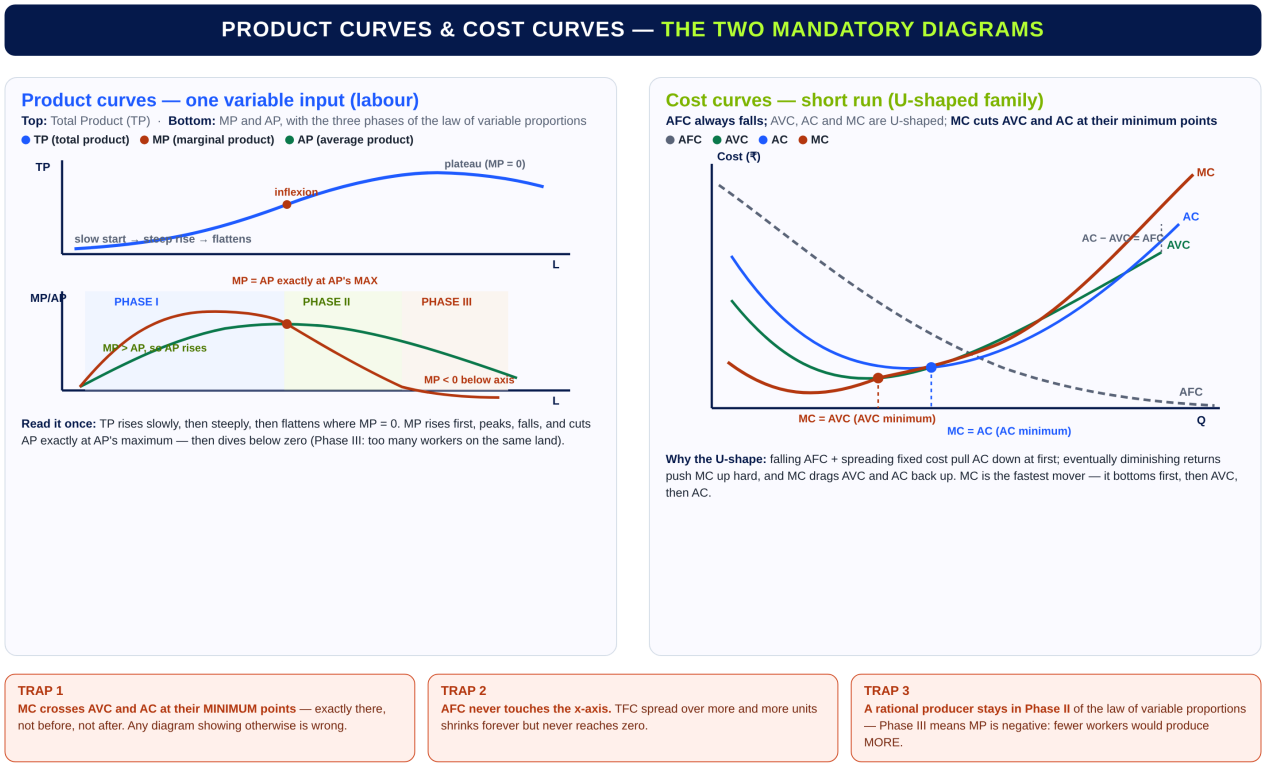


Figure 1: The two mandatory diagram families: product curves with the three phases (left) and the short-run cost family with MC cutting AVC and AC at their minima (right).

- **MP is the thermometer, AP follows it:** while MP > AP, AP rises; the moment MP < AP, AP falls; and MP crosses AP exactly at AP's peak. This single sentence answers most curve-diagram MCQs.

Returns to Scale: The Long-Run Story

Change ALL inputs together — double workers, ovens, flour, everything — and watch output. The long-run pattern is called **returns to scale**, and it has three faces.

Case	What happens	Everyday anchor
Increasing returns to scale	Output rises MORE than proportionally — inputs double, output triples	A bigger kitchen: two cooks + one big oven make more than double the small-oven output
Constant returns to scale	Output rises in the SAME proportion	Franchises: copying a shop design exactly copies its sales capacity
Decreasing returns to scale	Output rises LESS than proportionally	A giant bureaucracy: ten times the staff, far less than ten times the files cleared

- **Why increasing returns occur:** specialisation of labour, indivisible machines (one huge furnace beats ten small ones), and geometric advantages (a bigger tanker carries more than twice a small one).

- **Why decreasing returns occur:** management and coordination stretch thin — more meetings, more layers, more delays.
- **Do not mix the two laws:** variable proportions is SHORT RUN (one factor fixed); returns to scale is LONG RUN (all factors change together). The MCQ lives exactly in this distinction.

The Cost Family: Fixed, Variable, and the U-Shape

Every producer pays two kinds of bills. Rent continues even when output is ZERO; flour bills vanish when you bake nothing. That split builds the entire short-run cost family — and the family's portrait (the U-shapes) is the most-drawn diagram of this unit.

- **TFC (Total Fixed Cost):** independent of output — rent, insurance, permanent salaries. Same at 0 units and 6,000 units.
- **TVC (Total Variable Cost):** moves with output — materials, casual labour, power. Zero at zero output.
- **TC (Total Cost):** TC = TFC + TVC — the TC curve sits exactly TFC above the TVC curve and runs parallel to it.
- **The four per-unit cousins:**
  - AFC = TFC $/Q$ — falls FOREVER, sliding toward (never touching) the x-axis: the same fixed cost spread over more units.
  - AVC = TVC $/Q$ — U-shaped.
  - AC = TC $/Q$ = AFC + AVC — U-shaped, always above AVC by exactly the AFC gap.
  - MC = $\frac{\Delta \text{ TC}}{\Delta Q}$ — the extra cost of ONE more unit; the fastest-moving curve of the family.

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BUILD THE FULL COST TABLE — TFC = ₹60

One bakery, fixed cost ₹60 per day. TVC is given; compute everything else. Read each column's recipe from the formulas above.

Q	TFC	TVC	TC	AFC	AVC	AC	MC
0	60	0	60	—	—	—	—
1	60	20	80	60	20	80	20
2	60	30	90	30	15	45	10
3	60	45	105	20	15	35	15
4	60	70	130	15	17.5	32.5	25
5	60	105	165	12	21	33	35
6	60	150	210	10	25	35	45

- **TC column check:** 60 + 20 = 80 ✓ · 60 + 30 = 90 ✓ — every TC is TFC + TVC.

- **MC column check:** TC(2) − TC(1) = 90 − 80 = 10 ✓ — MC is the gap between two TCs.
- **Watch the U appear:** AC falls 80 → 45 → 35 → 32.5, bottoms at the 4th unit, then climbs 33 → 35. And MC falls first (20 → 10 → 15), crosses AVC exactly at its minimum (15 = 15 at Q3), and crosses AC at its minimum (between Q4 and Q5, where 25 < 32.5 then 35 > 33).

- **Why the U-shape:** at first, spreading AFC plus specialisation pulls AC down; beyond some output, diminishing returns push MC up hard, and rising MC drags AVC and then AC back up. MC bottoms out FIRST, then AVC, then AC.
- **The two crossings to tattoo on your memory:** MC cuts AVC at AVC's minimum; MC cuts AC at AC's minimum. Not before, not after — every diagram question checks exactly this.

Revenue under Perfect Competition

A wheat farmer can sell any quantity at today's market price — but not a rupee above it. One farmer among a lakh is a **price taker**. This single fact makes the revenue side of this chapter surprisingly simple.

- **TR (Total Revenue):** TR = $P \times Q$ — price times quantity sold; a straight line rising from the origin.
- **AR (Average Revenue):** AR = TR $/Q = P$ — revenue per unit IS the price; the AR curve is the firm's demand curve.
- **MR (Marginal Revenue):** MR = $\frac{\Delta \text{ TR}}{\Delta Q}$ — the extra revenue from one more unit. Sell at a fixed price → every unit adds exactly P rupees: MR = P
- **The magic collapse:** for a perfectly competitive firm, AR = MR = P — one horizontal line at market price. The firm can sell 10 units or 10,000 at P, so no unit ever sells for less than the previous one.

Measure	Formula	Behaviour for a price taker
TR	$P \times Q$	Straight line through the origin, slope = P
AR	TR $/Q = P$	Horizontal line at P — the firm's demand curve
MR	$(\Delta$ TR$)/(\Delta Q)$	Same horizontal line: MR = AR = P

- **Contrast worth one line:** a monopoly or any price-maker faces a DOWNWARD-sloping AR and MR below it — because selling more forces the price down. The horizontal AR = MR pair is the signature of perfect competition; you will use it heavily in the next unit.

Producer's Equilibrium: The MC = MR Rule

Producers chase profit, and profit thinking is marginal thinking: make one more unit ONLY if it adds more to revenue than to cost. Equilibrium is where that logic says stop.

THE THREE CONDITIONS OF PRODUCER'S EQUILIBRIUM

1. MC = MR — the last unit adds exactly as much cost as revenue.

2. MC must be RISING at that point — beyond it MC > MR, before it MC < MR.

3. Price ≥ AVC (short-run shutdown guard) — if price cannot even cover variable cost, produce nothing.

- **Why condition 2 matters:** MC can equal MR at TWO outputs when MC is U-shaped. If MC is falling at the crossing, one more unit adds less cost than revenue — the producer should keep expanding. Only the crossing on the RISING branch is the true equilibrium.
- **Why condition 3 matters:** fixed cost is sunk in the short run. If price cannot cover even the variable cost, every extra unit deepens the loss — shut down and lose only the TFC.

FIND THE EQUILIBRIUM OUTPUT — PRICE ₹25

The firm sells at ₹25 (price taker, so MR = ₹25 at every output). MC schedule from its cost curves:

Output	MR (₹)	MC (₹)	Marginal decision
1	25	10	Produce — 15 surplus over cost
2	25	8	Produce — surplus grows
3	25	12	Produce — still ahead
4	25	20	Produce — last profitable step? not yet
5	25	25	STOP — MC = MR = 25, and MC is rising (20 → 25 → 32)
6	25	32	Never — adds ₹7 of loss
7	25	45	Never — worse still

- **Equilibrium output: 5 units.** Condition 1: MC = MR = 25 ✓ · Condition 2: MC is rising at the 5th unit ✓ · Condition 3: price ₹25 covers AVC (the firm keeps producing in the short run) ✓
- **Below 5 units the firm leaves money on the table; above 5, each unit eats profit. Five is the sweet spot.**

Key Formulas & Rules: One-Stop Revision

PRODUCTION AND COSTS — EVERY RULE ON ONE PAGE

Rule	Statement	When to use it
Marginal product	$\text{MP} = (\Delta\ \text{TP})/(\Delta L)$	Extra output from one more worker
Average product	$\text{AP} = \text{TP}\ /L$	Output per worker
TP as sum	$\text{TP} = \text{MP}_1 + \text{MP}_2 + \cdots$	Rebuild TP from the MP column
AP maximum	MP = AP exactly there	Locate AP's peak in tables and diagrams
Total cost	$\text{TC} = \text{TFC} + \text{TVC}$	Split any cost bill
Average fixed cost	$\text{AFC} = \text{TFC}\ /Q$	Falls forever, never touches the axis
Average cost	$\text{AC} = \text{TC}\ /Q = \text{AFC} + \text{AVC}$	U-shaped; gap to AVC = AFC
Marginal cost	$\text{MC} = (\Delta\ \text{TC})/(\Delta Q)$	Cut-off rule for one more unit
MC crossings	MC = AVC at AVC's min; MC = AC at AC's min	Verify every cost diagram
Total revenue	$\text{TR} = P \times Q$	Straight line from the origin
Price taker	$\text{AR} = \text{MR} = P$	Horizontal line — perfect competition's signature
Producer equilibrium	MC = MR + MC rising + P ≥ AVC	All three, or the answer is bait
Phase II	MP positive but falling; AP falling	The rational producer's zone
Returns to scale	All inputs × 2 → output > 2× = IRS	Long-run law; never mix with variable proportions

Common Errors & Special Cases

THE TRAPS THIS UNIT SETS — READ BEFORE THE MOCK

The rule you assume	The twist	The trap in the question
"Long run means years"	It means ALL factors variable — duration is industry-specific	SR/LR classification MCQs
TP is max when MP is max	TP is max when MP = 0; TP falls once MP is negative	Reading TP/MP tables
MC can cut AC anywhere	MC cuts AC (and AVC) exactly at their MINIMA	Diagram-checking questions
AFC hits zero at high Q	AFC only approaches the axis — never touches it	AFC shape questions
Fixed cost exists in the long run	In the long run NOTHING is fixed — no TFC, no AFC	SR vs LR cost curve MCQs
MC = MR is enough for equilibrium	MC must also be RISING; price must cover AVC	Two-crossing MC diagrams
Returns to scale is about one factor	It is about ALL factors changing together	Mixing variable proportions with scale
MR falls like AR for all firms	For a price taker MR = AR = P — flat	Perfect competition revenue rows
AVC = AC always	AC = AVC + AFC; the gap (AFC) shrinks but never closes	Which-curve-is-on-top questions

The 60-Second Self-Test

Cover the answers, say each one aloud, then check. If any answer takes longer than ten seconds, revisit that section.

- **TP at 9 workers is 120 kg; the 10th worker adds 9 kg.** MP and AP of 10 workers? → MP = 9; AP = 129 ÷ 10 = 12.9.
- **TFC = ₹100 and Q = 4.** AFC? → 100 ÷ 4 = ₹25 — and at Q = 400 it would be ₹0.25, still not zero.
- **AC = ₹40 and AFC = ₹12.** AVC? → 40 − 12 = ₹28 — the gap between AC and AVC is always AFC.
- **At 6 units MC = 30 = MR and MC is falling.** Equilibrium? → NO — MC must be RISING; keep expanding past this false crossing.
- **Price ₹18, AVC at best output ₹22.** Produce? → No — price below AVC: shut down, lose only TFC.
- **All inputs double, output rises 2.6 times.** Which law, which run? → Increasing returns to scale — a LONG-RUN law.

The Whole Chapter in One Picture

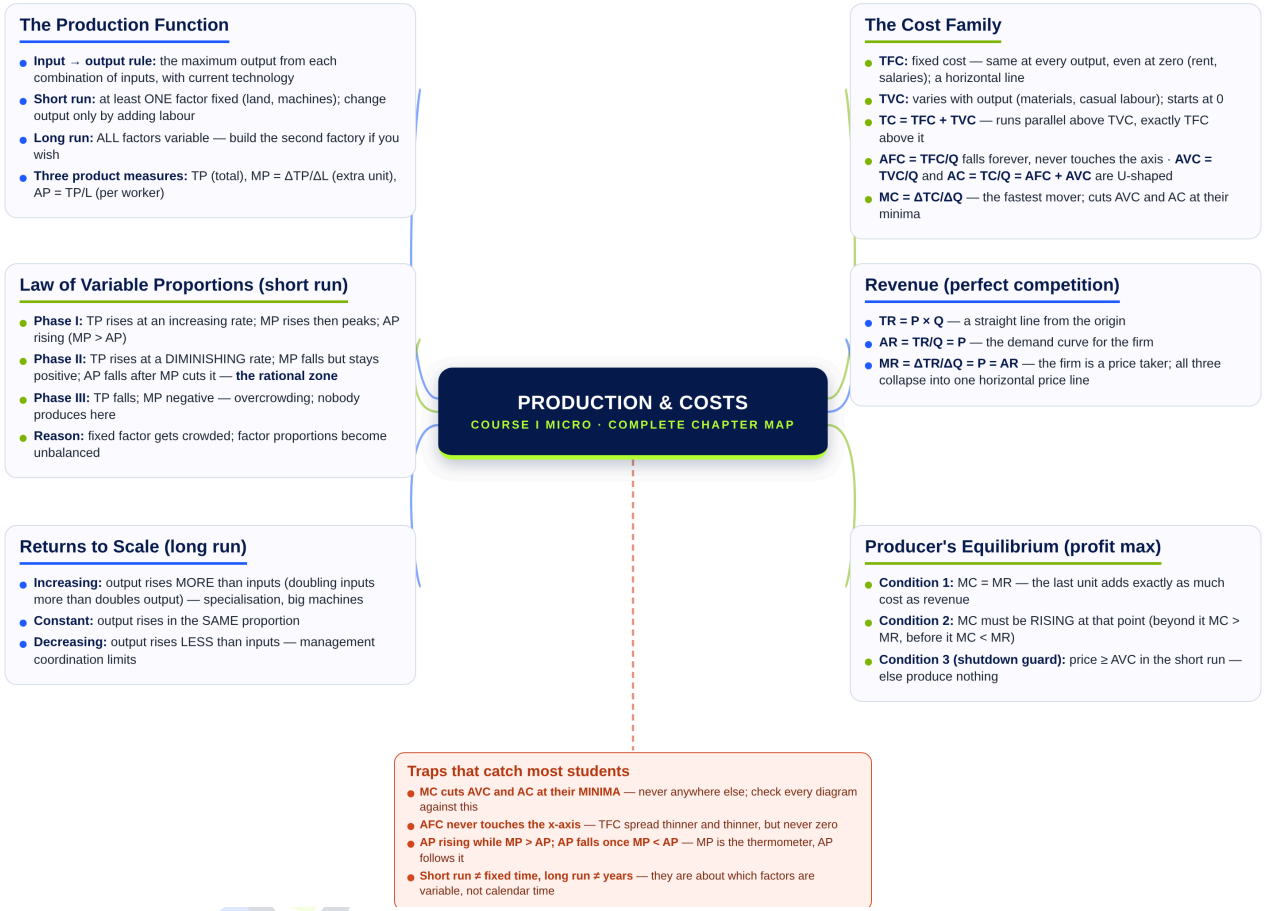


Figure 2: Chapter mindmap: the production function and its two runs, the law of variable proportions, returns to scale, the cost family, revenue under price-taking, and the three equilibrium conditions — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. The short run in production theory is a period in which:

(A) all factors are variable
(B) at least one factor is fixed
(C) exactly one year passes
(D) output cannot change at all

Q2. Total product rises from 120 units to 132 units when the 9th worker is hired. The marginal product of the 9th worker is:

(A) 13.3 units
(B) 12 units
(C) 132 units
(D) 9 units

Q3. In Phase II of the law of variable proportions:

(A) TP rises at an increasing rate
(B) MP is negative
(C) TP rises at a diminishing rate while MP stays positive
(D) AP rises throughout

Q4. A rational producer chooses to operate in:

(A) Phase I
(B) Phase II
(C) Phase III
(D) any phase, depending on price

Q5. A firm's total fixed cost is ₹60. At ZERO output, its total cost is:

(A) zero
(B) ₹60
(C) depends on AVC
(D) cannot be known

Q6. At 8 units of output, AFC is ₹12 and AVC is ₹20. AC equals:

(A) ₹32
(B) ₹8
(C) ₹160
(D) ₹28

Q7. Marginal cost cuts the average cost curve:

(A) at AC's maximum point
(B) at AC's minimum point
(C) where AC is still falling
(D) never — they run parallel

Q8. For a perfectly competitive firm, the AR and MR curves are:

(A) downward sloping, MR below AR
(B) the same horizontal line at the market price
(C) upward sloping
(D) U-shaped

Q9. A producer is in equilibrium only when MC = MR and:

(A) MC is falling
(B) MC is rising
(C) AC is at its minimum
(D) TR is at its maximum

Q10. A firm doubles ALL its inputs and output rises 2.6 times. This is a case of:

(A) diminishing returns to a factor
(B) increasing returns to scale
(C) negative returns to a factor
(D) constant returns to scale

Detailed Solutions

Each solution shows the working, not just the verdict — cover the answer line, attempt the reasoning, then compare.

Sol. Q1. The short run is defined by factor fixity, not by the clock: at least one input (land, plant, machines) cannot change, so output adjusts only through the variable factors. In the long run every factor is variable.

Answer: (B)

Sol. Q2. MP = change in TP: 132 − 120 = 12 units. The 9th worker added exactly 12 units to the total — averages (13.3) have no role here.

Answer: (B)

Sol. Q3. Phase II is the diminishing-returns zone: TP still rises (MP > 0) but by smaller steps each time, and AP falls after MP cuts it. MP turns negative only in Phase III.

Answer: (C)

Sol. Q4. Phase I wastes the fixed factor's capacity (MP still rising); Phase III destroys output (MP < 0). Only Phase II — MP positive but falling — is the rational zone.

Answer: (B)

Sol. Q5. Total cost = TFC + TVC. At zero output TVC is zero but TFC (₹60) is still owed — rent does not pause because you baked nothing. So TC = ₹60.

Answer: (B)

Sol. Q6. AC = AFC + AVC = 12 + 20 = ₹32. Cross-check: TC = 8 × 32 = ₹256; TFC = 8 × 12 = ₹96; TVC = 8 × 20 = ₹160; 96 + 160 = 256 ✓

Answer: (A)

Sol. Q7. MC is the fastest mover: below AC it pulls AC down, above AC it pushes AC up. So AC falls until MC meets it, and rises after — they must cross exactly at AC's minimum. Same logic pins MC to AVC's minimum.

Answer: (B)

Sol. Q8. A price taker sells every unit at the same market price, so revenue per unit (AR) and extra revenue per unit (MR) both equal P — one horizontal line. Downward-sloping AR with MR below it belongs to price-makers like a monopolist.

Answer: (B)

Sol. Q9. MC = MR has two crossings on a U-shaped MC. Only where MC is RISING does the next unit begin to cost more than it earns — the true stop point. A falling-MC crossing invites expansion, so it cannot be equilibrium.

Answer: (B)

Sol. Q10. Returns to scale is the LONG-RUN law about ALL inputs changing together. Inputs doubled (×2), output rose ×2.6 — more than proportionally: increasing returns to scale. "Returns to a factor" needs one fixed factor — not this case.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option C	Treating the run as time	SR/LR is about factor fixity — a tea stall's long run is a week, an airline's is a decade
Q2 option A	Computing AP instead of MP	MP = ΔTP only: 132 − 120 = 12; 132 ÷ 9 would be AP
Q5 option A	Forgetting the sunk rent	TFC is owed even at zero output — TC(0) = TFC, never zero
Q6 option D	Adding only AVC	AC = AFC + AVC = 12 + 20 = 32 — never drop the fixed cousin
Q8 option A	Importing monopoly shapes	Perfect competition's AR = MR = P — one flat line at market price
Q9 option C	Confusing AC-min with equilibrium	Equilibrium compares MARGINAL cost with MARGINAL revenue — AC's minimum is a cost-curve event, not the stop rule
Q10 option A	Mixing the two laws	One fixed factor → variable proportions; ALL factors doubled → returns to scale (here: increasing)

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